copies of what's on Mars, and all experienced from the safety of a virtual reality salon in your hometown. This is not *why* we explore Mars, but clearly we will need robot explorers to return the real reality before it can be reconfigured into virtual reality.

Especially with continuing investment in robotics and machine intelligence, sending humans to Mars can't be justified by science alone. And many more people can experience the virtual Mars than could possibly be sent to the real one. We can do very well with robots. If we're going to send people, we'll need a better reason than science and exploration.

In the 1980s, I thought I saw a coherent justification for human missions to Mars. I imagined the United States and the Soviet Union, the two Cold War rivals that had put our global civilization at risk, joining together in a far-seeing, high-technology endeavor that would give hope to people everywhere. I pictured a kind of *Apollo* program in reverse, in which cooperation, not competition, was the driving force, in which the two leading space-faring nations would together lay the groundwork for a major advance in human history —the eventual settlement of another planet.

The symbolism seemed so apt. The same technology that can propel apocalyptic weapons from continent to continent would enable the first human voyage to another planet. It was a choice of fitting mythic power: to embrace the planet named after, rather the madness ascribed to, the god of war.

We succeeded in interesting Soviet scientists and engineers in such a joint endeavor. Roald Sagdeev, then director of the Institute for Space Research of the Soviet Academy of Sciences in Moscow, was already deeply engaged in international cooperation on Soviet robotic missions to Venus, Mars, and Halley's Comet, long before the idea became fashionable. Projected joint use of the Soviet *Mir* space station and the *Saturn V*-class launch vehicle *Energiya* made cooperation attractive to the Soviet organizations that manufactured these items of hardware; they were otherwise having difficulty justifying their wares. Through a sequence of arguments (helping to bring the Cold War to an end being chief among them), then-Soviet leader Mikhail S. Gorbachev was convinced. During the December